

IoT Cloud Integration & Real-Time Remote Monitoring

Embedded Systems and IOT Task-4

Project Report on

SMART LIGHTING USING LDR WITH LCD, ESP32 AND BLYNK IOT DASHBOARD

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Submitted To



MainCrafts Technologies

Academic Year: 2025 – 2026

Project Domain:

Embedded Systems & Internet of Things (IoT)

Smart Lighting System Using LDR With LCD, Esp32 And Blynk IOT Dashboard

The Smart Lighting System using LDR with Lcd display is an advancement to the smart lighting system with Lcd (previous task project). It is an automated lighting solution designed to control a light source based on ambient light intensity. The system uses a ESP32 microcontroller to monitor light intensity through an LDR sensor and display the values and state of the light bulb. It provides the user to know about the state of light and intensity of light using an LCD display integrated with i2c and provide real time data to the blynk iot dashboard.

Components Required

Component Name	Quantity	Purpose
Esp32	1	Main microcontroller used to process sensor data and control the LED with WIFI connectivity.
Light Dependent Resistor (LDR)	1	Detects ambient light intensity.
10 kΩ Resistor	1	Forms a voltage divider circuit with the LDR.
LED	1	Represents the lighting system output.

220 Ω Resistor	1	Limits current through the LED and protects it from damage.
Lcd display with i2c	1	Used to display the values and state of the system.
Breadboard	1	Used for temporary circuit connections.

Working Principle

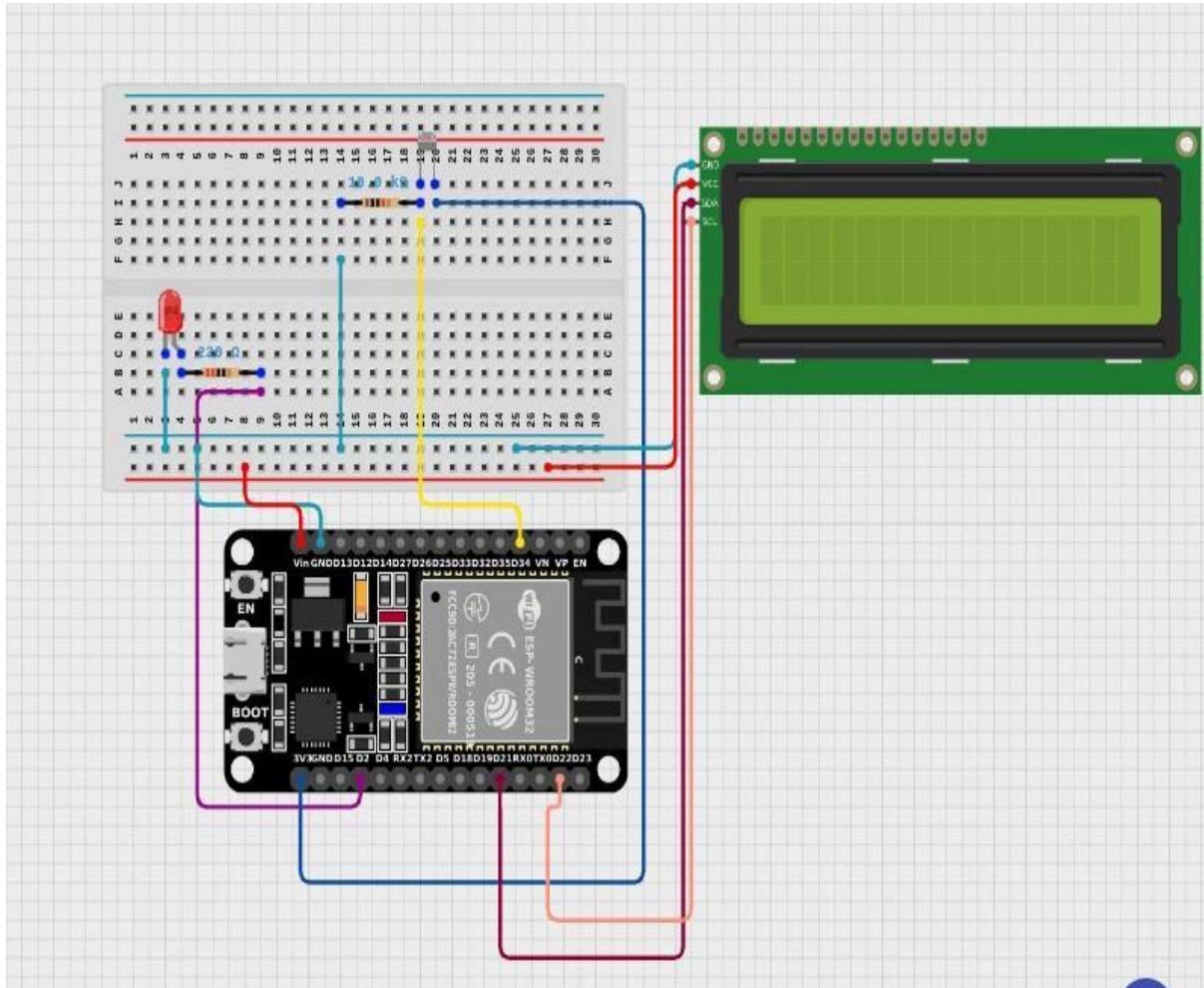
The Smart Lighting System using LDR with LCD display is a real life application in day to day life. It operates based on the intensity of ambient light detected by a Light Dependent Resistor (LDR). The resistance of the LDR varies according to the amount of light falling on its surface. When the surrounding environment is bright, the resistance of the LDR decreases, resulting in a higher voltage. When the environment becomes dark, the resistance increases, resulting in a lower voltage.

The LDR is connected to the Arduino Uno through a voltage divider circuit with 4.7k ohm resistor. The Arduino continuously reads the analog voltage from the LDR using its analog input pin (A0). The obtained value is compared with a threshold value programmed in the Arduino.

If the measured light intensity is below the threshold value, indicating low-light or dark conditions, the Arduino activates the output pin and turns the LED on connected at pin 4. When the light intensity rises above the threshold value, indicating sufficient brightness, the Arduino deactivates the output pin and turns the LED off.

This automatic process continuously repeats for every 500 milliseconds, allowing the system to respond to changes in environmental lighting conditions without human intervention. The system helps conserve energy, improves convenience, and provides an efficient solution for automatic lighting control in real life.

Circuit Diagram



Circuit diagram of smart lighting using LDR with LCD, ESP32, and blynk iot dashboard

Arduino IDE Code:

```
// SMART LIGHTING USING LDR WITH ESP32, LCD AND BLYNK IOT
```

```
#define BLYNK_TEMPLATE_ID "YOUR_TEMPLATE_ID"
```

```
#define BLYNK_TEMPLATE_NAME "YOUR_TEMPLATE_NAME"
```

```
#define BLYNK_AUTH_TOKEN "YOUR_AUTH_TOKEN"
```

```
#include <WiFi.h>
```

```
#include <WiFiClient.h>
```

```
#include <BlynkSimpleEsp32.h>
```

```
#include <Wire.h>
```

```
#include <LiquidCrystal_I2C.h>
```

```
char ssid[] = "YOUR WIFI NAME";
```

```
char pass[] = "YOUR WIFI PASSWORD";
```

```
#define LDR_PIN 34    // LDR sensor connected to Analog pin 34
```

```
#define LED_PIN 2     // LED connected to GPIO pin 2
```

```
int threshold = 1000; // Light threshold (adjust based on your environment)
```

```
LiquidCrystal_I2C lcd(0x27, 16, 2); // LCD I2C address

BlynkTimer timer;

// Function to read sensor data and update LCD & Blynk
void sendSensorData()
{
    int ldrValue = analogRead(LDR_PIN); // Read ADC value from LDR (0
to 4095)

    // Send ADC value to Blynk Virtual Pin V1
    Blynk.virtualWrite(V1, ldrValue);

    // Display ADC value on LCD
    lcd.setCursor(0, 0);
    lcd.print("LDR:");
    lcd.print(ldrValue);
    lcd.print("  ");

    // Print ADC value in Serial Monitor
    Serial.print("LDR VALUE: ");
    Serial.println(ldrValue);

    // if-else block

    // If it is dark (sensor value drops below threshold)
```

```
if (ldrValue < threshold)
{
    digitalWrite(LED_PIN, HIGH); // Turn ON light
    lcd.setCursor(0, 1);
    lcd.print("LIGHT ON   "); // Display status on LCD

    // Send light status to Blynk Virtual Pin V2
    Blynk.virtualWrite(V2, 1);
    Serial.println("Light ON");
}

// If it is bright
else
{
    digitalWrite(LED_PIN, LOW); // Turn OFF light

    lcd.setCursor(0, 1);
    lcd.print("LIGHT OFF  "); // Display status on LCD
    // Send light status to Blynk Virtual Pin V2
    Blynk.virtualWrite(V2, 0);
    Serial.println("Light OFF");
}
}

void setup()
```

```
{  
  pinMode(LED_PIN, OUTPUT);  
  pinMode(LDR_PIN, INPUT);  
  
  Serial.begin(115200); // Start serial monitor  
  lcd.init(); // Initiating I2C LCD screen  
  lcd.backlight();  
  lcd.setCursor(0, 0);  
  lcd.print("SMART LIGHTING");  
  
  // Connect ESP32 to Wi-Fi and Blynk Cloud  
  Blynk.begin(BLYNK_AUTH_TOKEN, ssid, pass);  
  
  // Execute sendSensorData() every 1 second  
  timer.setInterval(1000L, sendSensorData);  
  delay(2000);  
  lcd.clear();  
}  
void loop()  
{  
  Blynk.run(); // Maintain connection with Blynk server  
  timer.run(); // Run timer events  
}
```


Code Explanation

The Liquid Crystal header file is included to enable communication with the 16×2 LCD display.

The LDR sensor is connected to analog pin 34 to measure ambient light intensity.

An LED or relay is connected to digital pin 2 for automatic light control.

A threshold value of 1000 is defined to distinguish between dark and bright conditions.

In the `setup()` function, the input and output pins are configured, serial communication is started, and the LCD is initialized.

The LCD displays "smart lighting" when the system powers on.

In the `loop()` function, the Arduino continuously reads the LDR value using the `analogRead()` function.

The sensor reading is compared with the threshold value using an if-else condition.

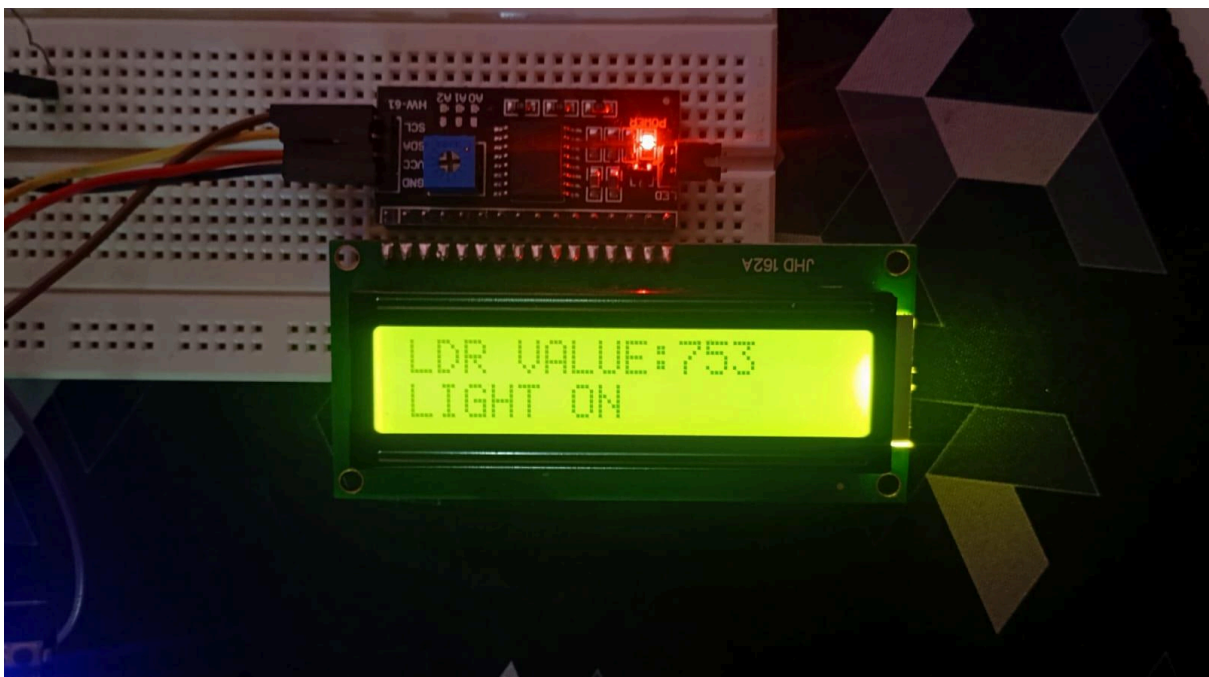
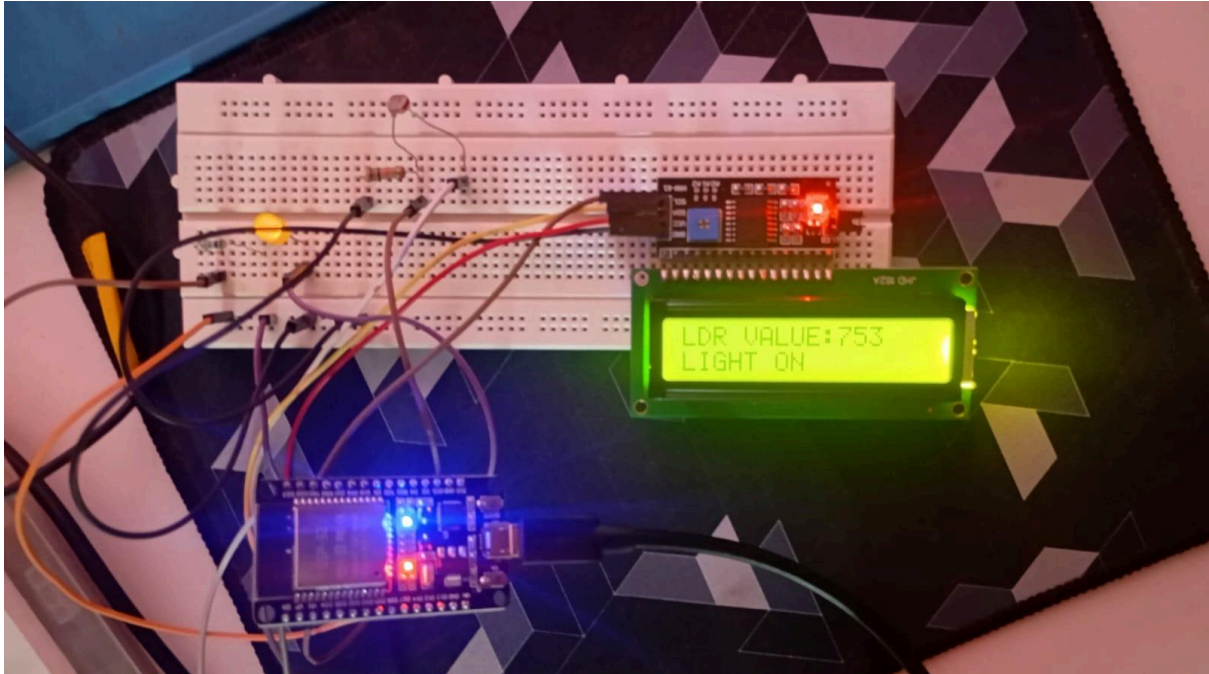
When the LDR value is below 1000, the environment is considered dark, and the LED is turned ON. The LCD displays "LIGHT ON".

When the LDR value is above or equal to 500, the environment is considered bright, and the LED is turned OFF. The LCD displays "LIGHT OFF".

The system continuously monitors light intensity and automatically controls the lighting, helping to save energy and improve convenience.

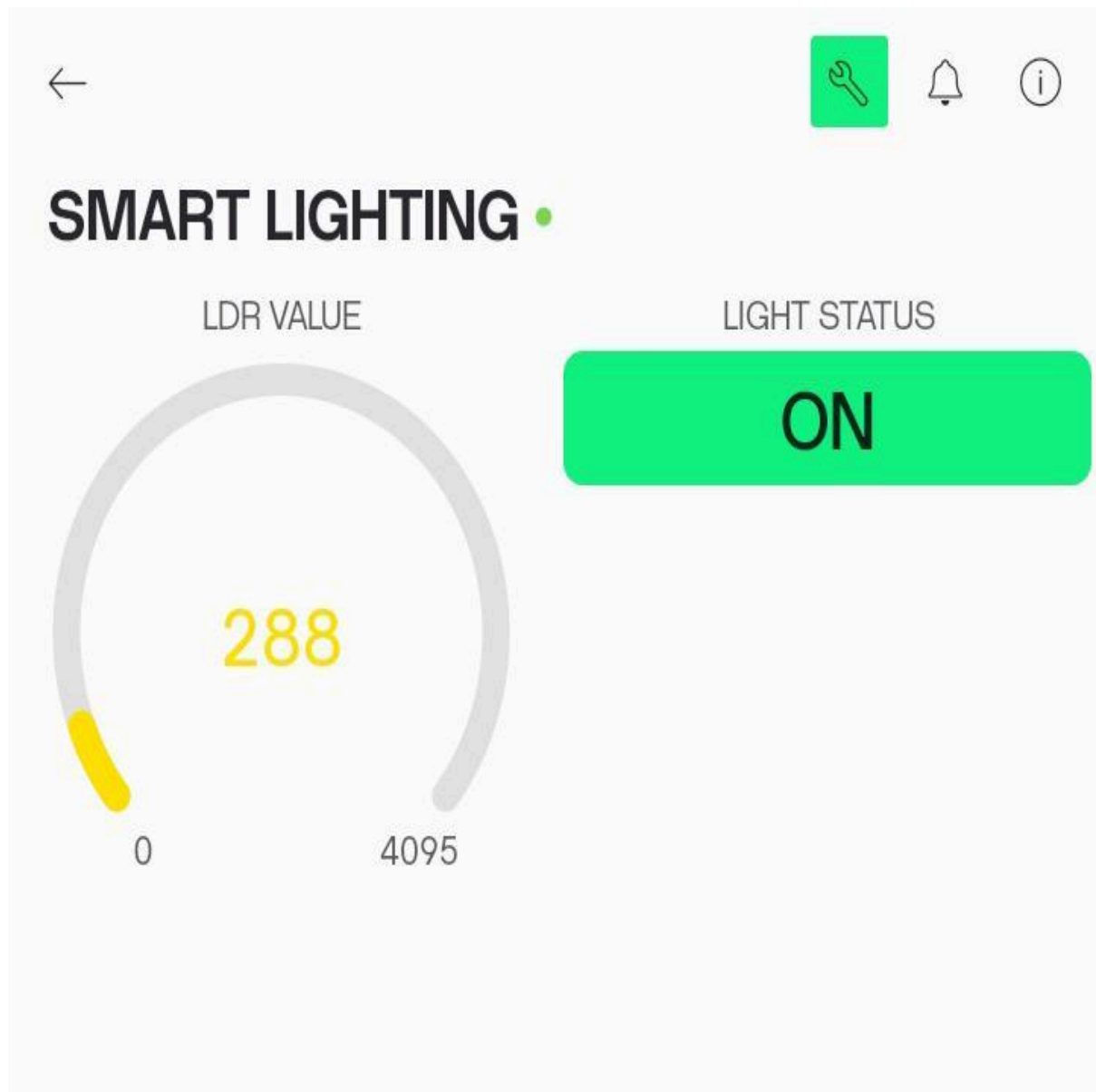
The system will update the Blynk iot dashboard through wifi and provide real time data to the user through mobile or desktop.

HARDWARE SETUP PICTURE

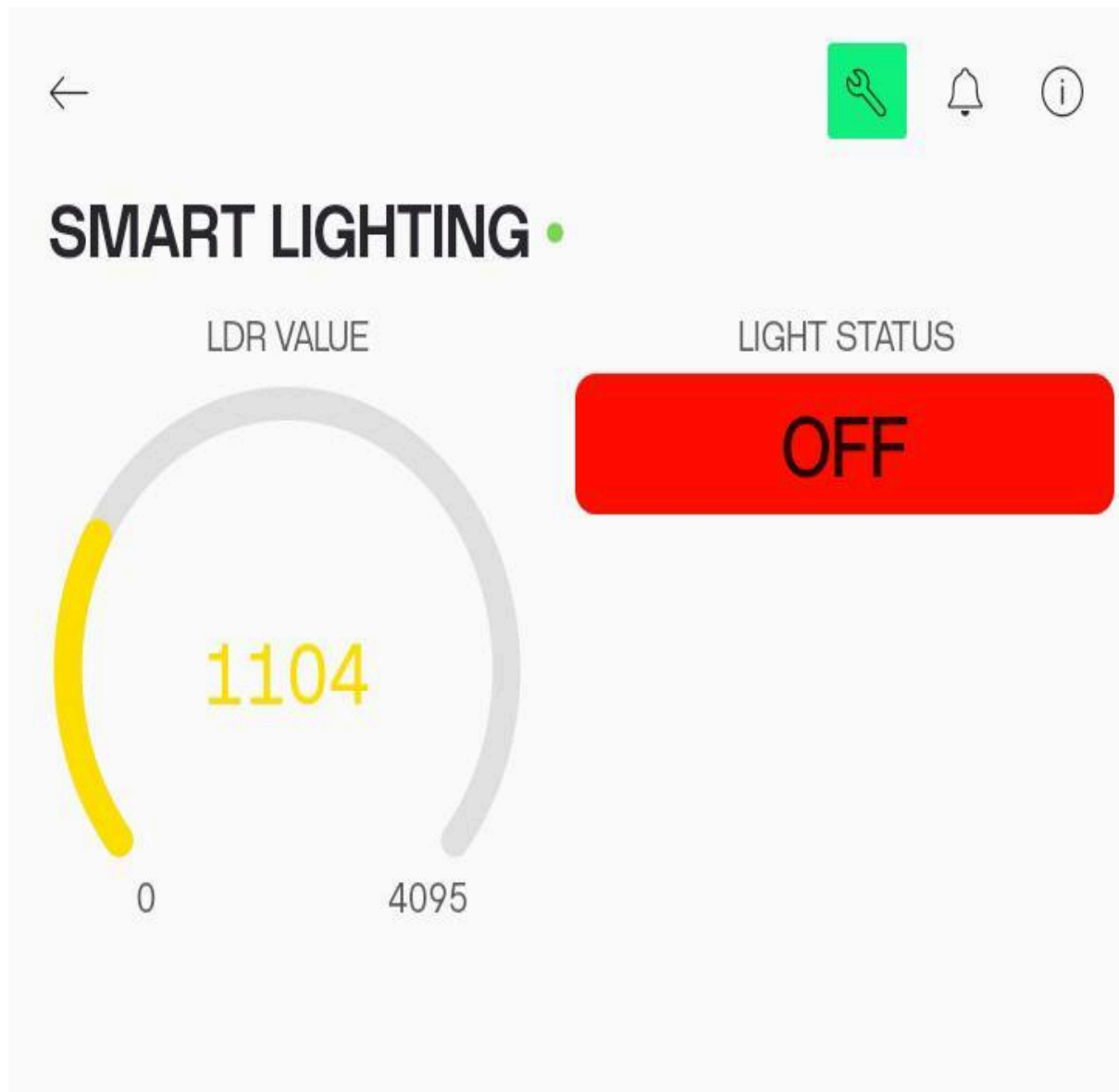


Output of lcd showing LIGHT ON when the brightness of light is lower than threshold value and turns the LED ON.

BLYNK IOT DASHBOARD SCREENSHOTS

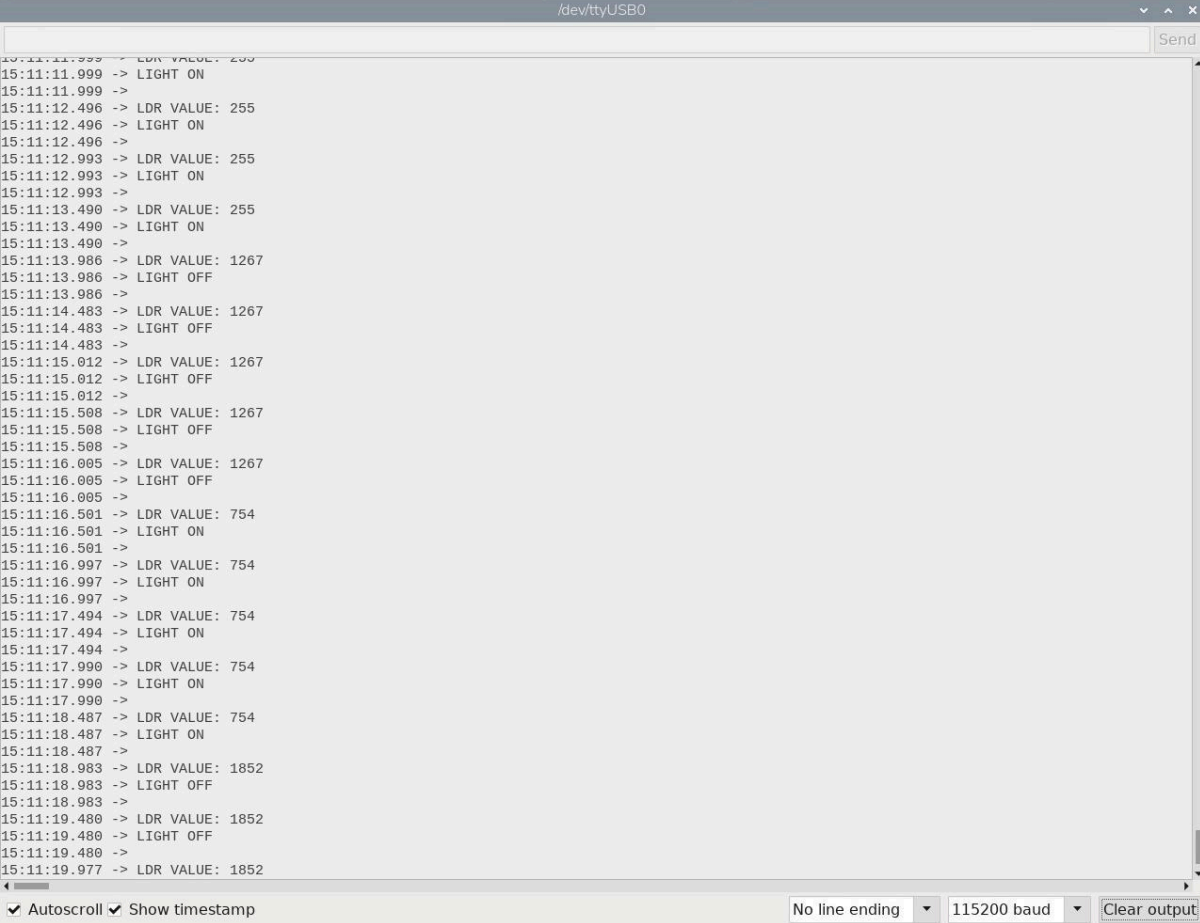


Output of IOT dashboard when the value of LDR is less than the threshold value and light status is ON.



Output of IOT dashboard when the value of IDR is more than the threshold value and light status is OFF.

SERIAL MONITOR OUTPUT:



The screenshot shows a serial monitor window titled "/dev/ttyUSB0" with a "Send" button in the top right corner. The main area displays a log of data received from the LDR circuit, including timestamps, LDR values, and light status. The status alternates between "LIGHT ON" and "LIGHT OFF" based on the LDR value. At the bottom, there are checkboxes for "Autoscroll" and "Show timestamp", and dropdown menus for "No line ending" and "115200 baud", along with a "Clear output" button.

```
15:11:11.999 -> LDR VALUE: 255
15:11:11.999 -> LIGHT ON
15:11:11.999 ->
15:11:12.496 -> LDR VALUE: 255
15:11:12.496 -> LIGHT ON
15:11:12.496 ->
15:11:12.993 -> LDR VALUE: 255
15:11:12.993 -> LIGHT ON
15:11:12.993 ->
15:11:13.490 -> LDR VALUE: 255
15:11:13.490 -> LIGHT ON
15:11:13.490 ->
15:11:13.986 -> LDR VALUE: 1267
15:11:13.986 -> LIGHT OFF
15:11:13.986 ->
15:11:14.483 -> LDR VALUE: 1267
15:11:14.483 -> LIGHT OFF
15:11:14.483 ->
15:11:15.012 -> LDR VALUE: 1267
15:11:15.012 -> LIGHT OFF
15:11:15.012 ->
15:11:15.508 -> LDR VALUE: 1267
15:11:15.508 -> LIGHT OFF
15:11:15.508 ->
15:11:16.005 -> LDR VALUE: 1267
15:11:16.005 -> LIGHT OFF
15:11:16.005 ->
15:11:16.501 -> LDR VALUE: 754
15:11:16.501 -> LIGHT ON
15:11:16.501 ->
15:11:16.997 -> LDR VALUE: 754
15:11:16.997 -> LIGHT ON
15:11:16.997 ->
15:11:17.494 -> LDR VALUE: 754
15:11:17.494 -> LIGHT ON
15:11:17.494 ->
15:11:17.990 -> LDR VALUE: 754
15:11:17.990 -> LIGHT ON
15:11:17.990 ->
15:11:18.487 -> LDR VALUE: 754
15:11:18.487 -> LIGHT ON
15:11:18.487 ->
15:11:18.983 -> LDR VALUE: 1852
15:11:18.983 -> LIGHT OFF
15:11:18.983 ->
15:11:19.480 -> LDR VALUE: 1852
15:11:19.480 -> LIGHT OFF
15:11:19.480 ->
15:11:19.977 -> LDR VALUE: 1852
```

☒ Autoscroll ☒ Show timestamp No line ending 115200 baud Clear output

Output of serial monitor printing the values of LDR circuit.